IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant / Inventor:	Reinaldo Javier Aguiar Marcano
Application Number:	19/561,223
Filing Date:	March 9, 2026
Confirmation Number:	1850
Art Unit / Examiner:	TC 2100 / ANDREA BENNETT
Office of Petitions:	Alesia M. Brown, Attorney Advisor
Customer Number:	226458
Entity Status:	SMALL ENTITY (37 CFR § 1.27)
Title of Invention:	SYSTEM AND METHOD FOR N-GRAM PARALLEL DATA ENCODING

SUPERSEDING AND SUPPLEMENTAL APPLICANT RESPONSE TO NOTICE OF INCOMPLETE REPLY AND FULL STATUTORY FEE REMITTANCE
(COMPLETE CURE OF ALL FEE DEFICIENCIES, FIVE-MONTH EXTENSION, CONDITIONAL SUSPENSION, AND FORMAL REFUND REQUEST)

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In responsive reply to the *Notice of Incomplete Reply* mailed August 5, 2026, the *Notice to File Missing Parts* mailed April 7, 2026, and the *Decision on Petition* (PETDEC) mailed October 1, 2026, and subsequent to the initial formalities response filed on October 10, 2026 (Electronic Receipt No. 82979426), the pro se Applicant, Reinaldo Javier Aguiar Marcano, respectfully submits this Superseding and Supplemental Response and full statutory fee remittance.

I. SUPERSEDING STATUS & RELATIONSHIP TO OCTOBER 10, 2026 SUBMISSION

This superseding and supplemental instrument is executed and submitted on October 11, 2026, subsequent to the filing of October 10, 2026. To eliminate any clerical ambiguity or procedural

uncertainty during pre-examination intake processing on Tuesday, October 13, 2026, and to ensure that all statutory fees are 100% funded in real-time, Applicant explicitly directs that **this October 11, 2026 instrument supersedes the fee transmittal and deposit allocation requests of October 10, 2026 in their entirety**.

The accompanying concurrent electronic payment in Patent Center completely satisfies all statutory fee requirements in full, rendering this response complete, controlling, and self-contained upon intake.

II. STATEMENT OF REAFFIRMED ENTITY STATUS (37 CFR § 1.27)

Applicant hereby formally reaffirms entitlement to **Small Entity status** under 37 CFR § 1.27 and 35 U.S.C. § 123. Applicant's verified Statement of Small Entity Status was filed on August 1, 2026, and officially recognized by the Office. All statutory fees are governed strictly by the Small Entity fee schedule.

III. PROCEDURAL POSTURE & CURED FORMALITIES STATUS

Applicant respectfully notes the following verified milestones established in the Image File Wrapper:

1. **Substitute Specification (37 CFR § 1.125; Rules R-01, R-05, R-14):** Concurrently herewith, Applicant submits a clean Substitute Specification in full compliance with 37 CFR § 1.125(c) formatted in strict double line spacing pursuant to 37 CFR § 1.52(b)(2)(i), excising all embedded drawing figures (satisfying 37 CFR § 1.58(a)), embedded claims, and embedded abstract text, and resolving mathematical typography (N^2). In further compliance with 37 CFR § 1.125(c), Applicant concurrently submits a separate marked-up copy of the substitute specification showing all changes relative to the as-filed specification.

2. **Formal Drawing Sheets (37 CFR § 1.84, § 1.121(d); Rules R-14, R-30):** Concurrently herewith, Applicant submits formal standalone drawing sheets comprising Sheet 1 of 2 (FIG. 1) and Sheet 2 of 2 (FIG. 2) presented on plain white paper, properly labeled in Arabic numerals, and containing standardized top margin indicia and the "Replacement Sheet" identification pursuant to 37 CFR § 1.121(d).

3. **Perfected Replacement Claims (37 CFR § 1.75(h), § 1.121):** Concurrently herewith, Applicant submits a clean replacement claim listing for Claims 1–6 commencing on a separate

sheet, formatted in strict double line spacing pursuant to 37 CFR § 1.52(b)(2)(i), consecutively numbered in Arabic numerals pursuant to 37 CFR § 1.126 under the pure statutory heading "CLAIMS" and introductory phrase "What is claimed is:". Crucially, this perfected claim listing resolves mathematical typography in Claim 4, correctly rendering exponential quantities as 2^{24} states and 2^{48} states (permanently curing LaTeX circumflex accent artifacts from prior papers), and sets forth Claim 1 in standard indented sub-paragraphs without bullet points.

4. **Perfected Replacement Abstract (37 CFR § 1.72(b), § 1.121):** Concurrently herewith, Applicant submits a clean replacement Abstract commencing on a separate sheet, formatted in strict double line spacing under the pure statutory heading "ABSTRACT OF THE DISCLOSURE", set forth as a single narrative paragraph of 114 words (under the 150-word statutory ceiling) with standard typographical quotation marks.

5. **Perfected AIA Inventor's Declaration (37 CFR § 1.63; Form PTO/AIA/01; Rule R-38):** Concurrently herewith, Applicant submits a perfected, fully compliant Inventor's Oath or Declaration under 37 CFR § 1.63 and MPEP § 605, explicitly setting forth: (a) the unified statutory authorization and original inventorship averment under 37 CFR § 1.63(a)(1), (4); (b) full acknowledgment of the duty of disclosure under 37 CFR § 1.56; (c) the inventor's complete legal name (*Reinaldo Javier Aguiar Marcano*), legal citizenship (*United States of America*), complete physical legal residence street address (*26714 Valleyside Drive, Katy, Texas 77494, USA*), and distinct mailing address (*2302 Britton Ridge Drive, Katy, Texas 77494, USA*) pursuant to 37 CFR § 1.63(b); (d) criminal false statement warning under 18 U.S.C. § 1001; and (e) compliant S-signature, permanently curing any latent informality from the initial July 21, 2026 declaration.

6. **Technical Sheets Intact:** All technical papers and inventor declarations required to place this application in condition for examination are fully satisfied and on file.

A. Formal Certification of No New Matter (37 CFR § 1.125(b) & MPEP § 608.01(q))

Pursuant to 37 CFR § 1.125(b) and MPEP § 608.01(q), Applicant hereby formally certifies that **the substitute specification submitted concurrently herewith includes no new matter**. The technical description corresponds verbatim to the original technical disclosure filed on March 9, 2026, subject only to the following formal corrections:

(a) **Excision of Embedded Drawings (37 CFR § 1.58(a)):** Complete excision of embedded graphical figures from original pages 5 and 6 into standalone formal drawing sheets (FIG. 1 and FIG. 2) under 37 CFR § 1.84;

(b) **Excision of Embedded Claims (37 CFR § 1.75(h)):** Complete excision of the claims from original pages 3 and 4 into a standalone replacement claim listing;

(c) **Excision of Embedded Abstract (37 CFR § 1.72(b)):** Complete excision of the abstract from original page 1 into a standalone replacement abstract sheet;

(d) **Line Spacing Compliance (37 CFR § 1.52(b)(2)(i)):** Formatting in strict double line spacing throughout; and

(e) **Mathematical Typographic Correction:** Typographical rendering of the state-space formula in the Summary of the Invention to express $N \times N = N^2$ in standard mathematical notation, curing the LaTeX circumflex artifact.

IV. COMPREHENSIVE CURATION OF ALL FEE DEFICIENCIES, CONCURRENT STATUTORY REMITTANCE, AND FORMAL REFUND REQUEST (37 CFR § 1.26)

A. Item-by-Item Reconciliation and Full Payment of All Required Pre-Examination Base Fees

In the August 5, 2026 Notice of Incomplete Reply, the Office identified specific statutory fee deficiencies under 37 CFR § 1.16. To ensure that this application is completely cured and immediately advanced to Technology Center 2100, Applicant concurrently remits payment in full for each and every statutory fee item identified by the Office:

1. **Basic Filing Fee (37 CFR § 1.16(a), Small Entity Fee Code 4011): $70.00**. Noted as unpaid in the August 5, 2026 Notice under "Items Required to Avoid Abandonment" ("The statutory basic filing fee has not been received"). Remitted in full concurrently herewith.
2. **Non-DOCX Format Penalty Fee (37 CFR § 1.16(u), Small Entity Fee Code 2054): $172.00**. Itemized in the August 5, 2026 Notice ("The fee set forth in 37 CFR 1.16(u) for submitting documents not in DOCX format has not been received... the specification was not submitted in DOCX format"). Remitted in full concurrently herewith.
3. **Application Search Fee (37 CFR § 1.16(k), Small Entity Fee Code 2111): $308.00**. Noted as unpaid in the August 5, 2026 Notice ("The application search fee has not been received"). Remitted in full concurrently herewith.
4. **Application Examination Fee (37 CFR § 1.16(o), Small Entity Fee Code 2311): $352.00**. Noted as unpaid in the August 5, 2026 Notice ("The application examination fee has not been received"). Remitted in full concurrently herewith.

5. **Late Surcharge Status (37 CFR § 1.16(f)):** The statutory late surcharge of **$68.00** under 37 CFR § 1.16(f) was previously remitted in full on July 31, 2026 (Fee Code 2051). Accordingly, zero surcharge fees remain due.

Subtotal Pre-Examination Base Fees Remitted/Satisfied: $902.00 gross Small Entity fees ($70 basic + $172 non-DOCX + $308 search + $352 exam), satisfying the $486.00 net balance itemized in the Notice in full.

B. Five-Month Extension of Time Fee Remittance (37 CFR § 1.17(a)(5))

In the August 5, 2026 Notice and the October 1, 2026 Decision on Petition, the Office noted that the extendable response period runs through November 9, 2026 upon payment of the fifth-month extension fee. Concurrently herewith, Applicant remits the entire statutory Small Entity fee:

- **Five-Month Extension of Time Fee (37 CFR § 1.17(a)(5), Fee Code 2255): $1,358.00**. Remitted in full via electronic fee payment in Patent Center, rendering this response indisputably timely under 35 U.S.C. § 133 and 37 CFR § 1.136(a) through Monday, November 9, 2026.

C. Suspension Processing Fee Remittance (37 CFR § 1.17(i))

To ensure immediate standing for the accompanying Request for Suspension under 37 CFR § 1.103(a), Applicant remits:

- **Processing Fee for Suspension Request (37 CFR § 1.17(i), Fee Code 2830): $60.00**. Remitted in full via electronic fee payment in Patent Center.

D. Complete Formalities and Fee Curation Statement

Applicant hereby expressly certifies that with the filing of the replacement claims, replacement abstract, the accompanying perfected Inventor's Declaration under 37 CFR § 1.63, and the concurrent electronic remittance of every statutory fee itemized above, **all formal deficiencies and all fee deficiencies cited in the April 7, 2026 Notice to File Missing Parts, the August 5, 2026 Notice of Incomplete Reply, and the October 1, 2026 Decision on Petition are 100% completely cured**, leaving zero unresolved defects, zero missing parts, and zero outstanding fee balances under 37 CFR § 1.16 or § 1.17.

E. Formal Request for Refund under 37 CFR § 1.26

Because Applicant is remitting all required statutory fees concurrently herewith via electronic fee payment in Patent Center, Applicant respectfully directs the Office of Finance to the accumulated unapplied funds and overlapping payments currently held under this application:

1. **$416.00 in Unapplied Deposit Funds:** Official Office records (Notice of Incomplete Reply mailed August 5, 2026 and Fee Worksheet WFEE) confirm that the Office of Finance holds $416.00 in unapplied deposit funds under this application serial number;
2. **$276.00 Overlapping Extension Fee:** Applicant previously remitted $276.00 on July 31, 2026 for a two-month extension of time (Fee Code 2252). Because Applicant is paying the full $1,358.00 five-month extension fee in full today, the $276.00 prior payment represents an overlapping overpayment.

Pursuant to 37 CFR § 1.26 and MPEP § 509, Applicant hereby respectfully requests that the Office of Finance process a refund of said accumulated credits totaling **$692.00** ($416.00 + $276.00) via Treasury disbursement payable to Applicant at the correspondence address of record, as all statutory obligations are fully satisfied by the concurrent fee remittance submitted herewith.

V. CONCLUSION

All formality requirements and statutory fees having been fully satisfied, Applicant respectfully requests that the Office of Patent Application Processing (OPAP) advance the application to Technology Center 2100, and that the Office enter and grant the accompanying Conditional Request for Suspension of Action under 37 CFR § 1.103(a) for a period of six (6) months.

Concurrently submitted herewith are the following discrete documents under their official Patent Center UI descriptions:

(1) **Applicant Response to Pre-Exam Formalities Notice** (This Transmittal Instrument)

(2) **Extension of Time** (Five-Month Extension Petition under 37 CFR § 1.136(a), Fee Code 2255)

(3) **Letter Requesting Suspension of Action** (Conditional Suspension Request under 37 CFR § 1.103(a), Fee Code 2830)

(4) **Oath or Declaration filed** (Perfected AIA Inventor Declaration under 37 CFR § 1.63, Rule R-38)

(5) **Specification** (Clean Substitute Specification under 37 CFR § 1.125(c))

(6) **Specification** (Marked-Up Copy of Substitute Specification under 37 CFR § 1.125(c))

(7) **Drawings** (Formal Drawing Sheets 1 and 2 under 37 CFR § 1.84 and § 1.121(d))

(8) **Claims** (Replacement Claims 1–6 under 37 CFR § 1.75(h) and § 1.121)

(9) **Abstract** (Replacement Abstract under 37 CFR § 1.72(b) and § 1.121)

Respectfully submitted,

/Reinaldo Javier Aguiar Marcano/
Reinaldo Javier Aguiar Marcano
Inventor and Pro Se Applicant
Mailing Address: 2302 Britton Ridge Drive, Katy, Texas 77494
Telephone: (202) 599-3979
Email: `rj.aguiar@gmail.com` (USPTO Account)
Email: `rj.aguiar.m@gmail.com`
Email: `reinaldo.aguiar.marcano@isbn9781971481296.com`
Date: October 11, 2026